

# Dataset report: PBMC3k single-cell RNA-seq

Source: Scanpy datasets.pbmc3k and pbmc3k\_processed; 10x PBMC3k

Dataset **pbmc3k** · run 2026-09-29 · job 2840720 · commit 766f1d17e2

Reporting mode: deterministic. This report was rendered from saved, validated evidence by deterministic templates; no language model wrote it. Every number below is read from a checksummed artifact in the evidence bundle.

## At a glance

|                      |                                                                                                                                |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Source               | 2,700 observations × 32,738 numerical features; the final analysis uses all 2,638 observations that pass QC (62 removed by QC) |
| Preprocessing branch | count matrix (chosen from the data profile, not the dataset name)                                                              |
| Pilot                | 1,000 observations; all ten methods; 3 seeds for stochastic methods; 5 refits on 80% subsamples for every method               |
| Retained             | PCA, MDS, t-SNE                                                                                                                |
| Final cohort         | 2,638 observations; binding constraint: all available observations                                                             |

## Findings

- On the pilot cohort, MDS has the highest pre-registered quality ( $Q = 0.567$ ); no other valid method falls within its equivalence margin.
- Among valid pilot methods, t-SNE has the highest neighbor recall (0.255).
- Among valid pilot methods, MDS has the highest global structure (0.855).
- Among valid pilot methods, Kernel PCA has the highest subsample stability (0.870).
- Severe diagnostics made these methods ineligible: Diffusion maps (collapsed layout (median collapse ratio  $0.009 < 0.05$  and  $< 0.25 \times \text{PCA } 0.253$ ), near-disconnected diffusion graph (Markov spectral gap  $9.0e-08$ )).
- MDS, t-SNE returned identical embeddings for every seed, so seed stability is trivially 1; the subsample stability used in  $Q$  is not affected.
- Across 10 rule variants, PCA is retained in every variant.
- On the final cohort ( $n = 2,638$ ), t-SNE has the highest trustworthiness (0.897), t-SNE the highest neighbor recall (0.208) and MDS the best global structure (0.842).
- The secondary label silhouette is highest for t-SNE (0.419); labels were not used for any fit or decision.
- The first two principal components explain 3.8% of the preprocessed feature variance; 50 components explain 14.1%.

## 1 · Dataset inspection

|                    |                                                                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage            | float32; sparsity 0.9741; values in [0, 419]; non-negative: True; integer-valued: True                                                                                   |
| Features           | 16104 constant, 16715 duplicated; feature variance min/median/max 0, 0.0003702, 2008; mitochondrial (MT-) features: 13                                                   |
| Observation totals | row sums min/median/max 548, 2197, 1.584e+04                                                                                                                             |
| Memory             | dense float64 674 MB; full pairwise matrix 56 MB (never materialized by the evaluation code)                                                                             |
| Secondary labels   | B cells 342 · CD14+ Monocytes 480 · CD4 T cells 1,144 · CD8 T cells 316 · Dendritic cells 37 · FCGR3A+ Monocytes 150 · Megakaryocytes 15 · NK cells 154 · unannotated 62 |

Labels only stratify cohort composition and color secondary plots. They never enter preprocessing fits, embeddings or selection.

## 2 · Preprocessing plan and rationale

Rule evidence: sparsity 0.974, non-negative True, integer-valued True, 32,738 features → branch **count matrix**. The same plan is re-fitted inside each cohort (pilot and final).

| Step                            | Parameters (final cohort)                                                           | Reason                                                                                                                                                                                                                                | Fit scope                        |
|---------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| observation qc                  | min_features=200;<br>max_features=2500;<br>max_mito_fraction=0.05;<br>retained=2638 | Count matrix (sparsity 0.974 $\geq$ 0.5, non-negative integer values, 32,738 features $\geq$ 1000): remove low-information and extreme observations with configured thresholds; 13 MT-features enable a mitochondrial-fraction filter | per-observation fixed thresholds |
| sample cohort                   | size=2638; seed=774;<br>stratified=True                                             | Common representative cohort; labels, when present, only preserve class proportions                                                                                                                                                   | cohort design                    |
| filter features                 | min_cells=3;<br>retained=13656                                                      | Remove features detected in too few cohort observations                                                                                                                                                                               | analysis cohort only             |
| normalize log1p                 | target_sum=1e+04                                                                    | Library sizes vary from 548 to 15,844 (median 2,197): normalize totals and log-compress counts                                                                                                                                        | analysis cohort only             |
| select highly variable features | n_top=2000; flavor=seurat;<br>n_selected=2000                                       | 32,738 sparse features: keep the most variable after normalization                                                                                                                                                                    | analysis cohort only             |
| scale clip                      | max_value=10                                                                        | Balance feature scales and limit outliers                                                                                                                                                                                             | analysis cohort only             |
| common pca representation       | n_components=50                                                                     | Common denoising and computational input for all ten methods; the displayed PCA is two-dimensional                                                                                                                                    | analysis cohort only             |

### 3 • Comparable pilot design

Every method was fitted to the identical saved input (1,000 × 50; SHA-256 261e8a8e98ce...). Stochastic methods (MDS, GPLVM, t-SNE, UMAP) use 3 seeds. Every method is also refitted on the same 5 random 80% subsamples. Metrics use Euclidean distance in the common input space:

- **T**: trustworthiness at  $k = 15$  (scikit-learn definition).
- **R**: mean k-nearest-neighbor recall, self excluded, stable tie order.
- **G**: Spearman correlation of pairwise distances (all pairs up to 500,000, otherwise a fixed random sample), clipped at 0.
- **S**: mean kNN overlap of subsample refits on their shared observations.

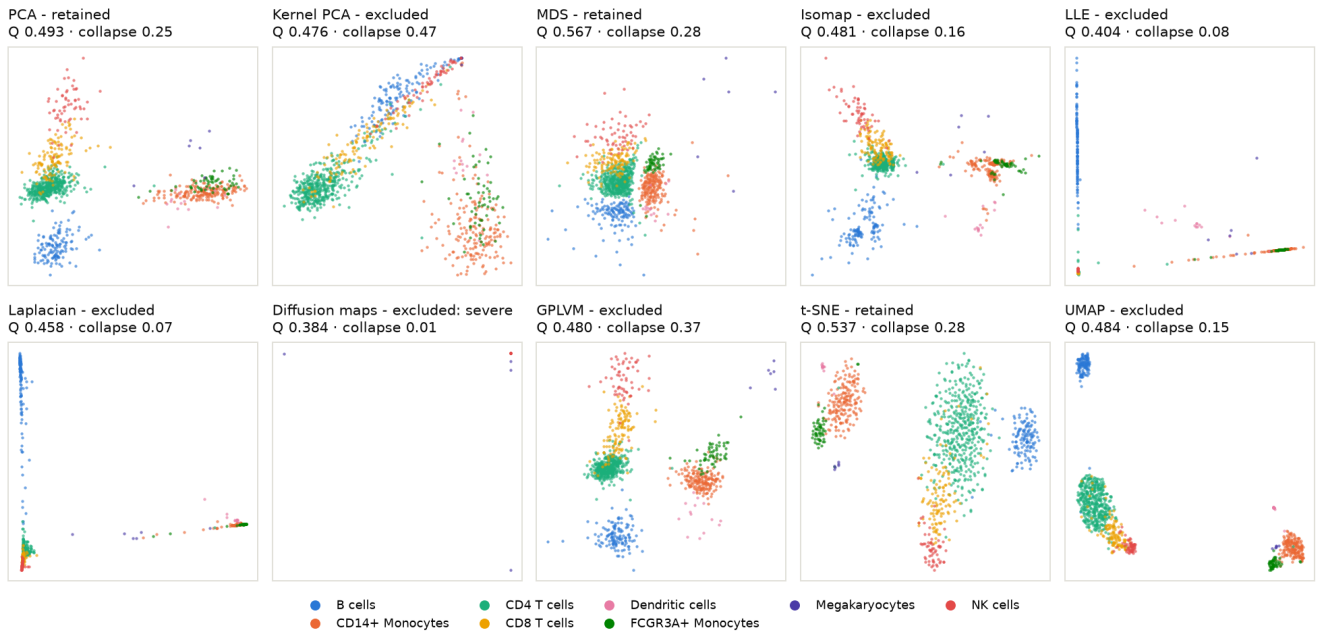
**Quality Q = 0.35 trustworthiness + 0.35 neighbor recall + 0.2 global structure + 0.1 stability.** The weights and all thresholds were fixed before the real-data run (docs/METHOD\_NOTES.md). Runtime is fit\_transform time only; imports and JIT warm-up are recorded separately.

Severe diagnostics make a method ineligible: disconnected-graph or singular warnings; a collapsed layout (median collapse ratio  $< 0.05$  and  $< 0.25 \times$  PCA's); a diffusion Markov spectral gap  $< 1e-06$ ; SMACOF reaching max\_iter.

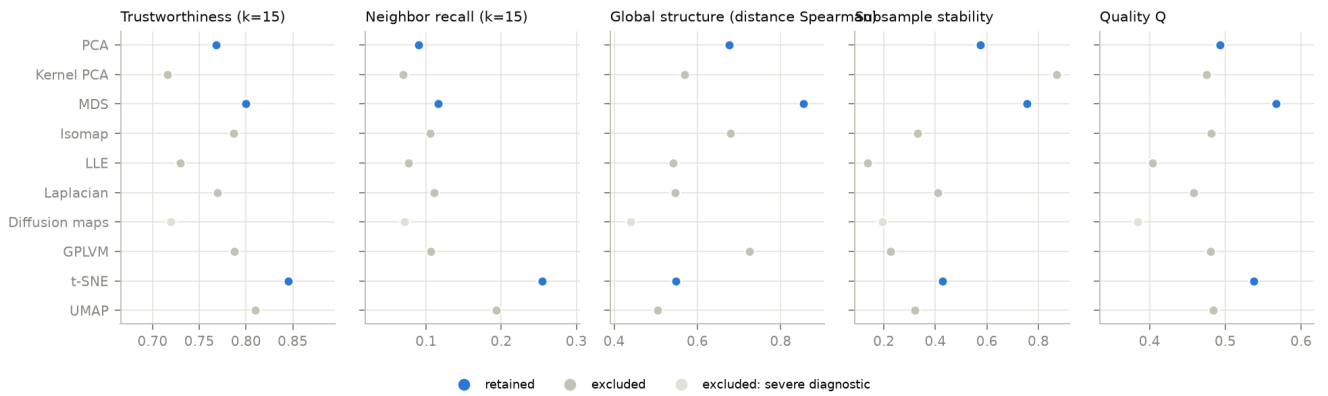
### 4 • Pilot results

| Method         | Status  | T     | R     | G     | S     | Seed S | Q     | Fit s | Collapse | Flags                                                                                                                                                     |
|----------------|---------|-------|-------|-------|-------|--------|-------|-------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| PCA            | success | 0.768 | 0.090 | 0.677 | 0.574 | -      | 0.493 | 0.00  | 0.25     |                                                                                                                                                           |
| Kernel PCA     | success | 0.716 | 0.069 | 0.569 | 0.870 | -      | 0.476 | 0.03  | 0.47     |                                                                                                                                                           |
| MDS            | success | 0.800 | 0.116 | 0.855 | 0.756 | 1.000  | 0.567 | 4.14  | 0.28     | seed-invariant                                                                                                                                            |
| Isomap         | success | 0.787 | 0.105 | 0.680 | 0.331 | -      | 0.481 | 0.28  | 0.16     |                                                                                                                                                           |
| LLE            | success | 0.730 | 0.077 | 0.541 | 0.137 | -      | 0.404 | 0.21  | 0.08     |                                                                                                                                                           |
| Laplacian      | success | 0.769 | 0.111 | 0.546 | 0.411 | -      | 0.458 | 0.10  | 0.07     |                                                                                                                                                           |
| Diffusion maps | success | 0.720 | 0.071 | 0.440 | 0.194 | -      | 0.384 | 0.36  | 0.01     | collapsed layout (median collapse ratio $0.009 < 0.05$ and $< 0.25 \times$ PCA 0.253); near-disconnected diffusion graph (Markov spectral gap $9.0e-08$ ) |
| GPLVM          | success | 0.787 | 0.106 | 0.725 | 0.225 | 0.164  | 0.480 | 9.58  | 0.37     |                                                                                                                                                           |
| t-SNE          | success | 0.845 | 0.255 | 0.548 | 0.428 | 1.000  | 0.537 | 6.01  | 0.28     | seed-invariant                                                                                                                                            |
| UMAP           | success | 0.810 | 0.193 | 0.505 | 0.320 | 0.321  | 0.484 | 2.02  | 0.15     |                                                                                                                                                           |

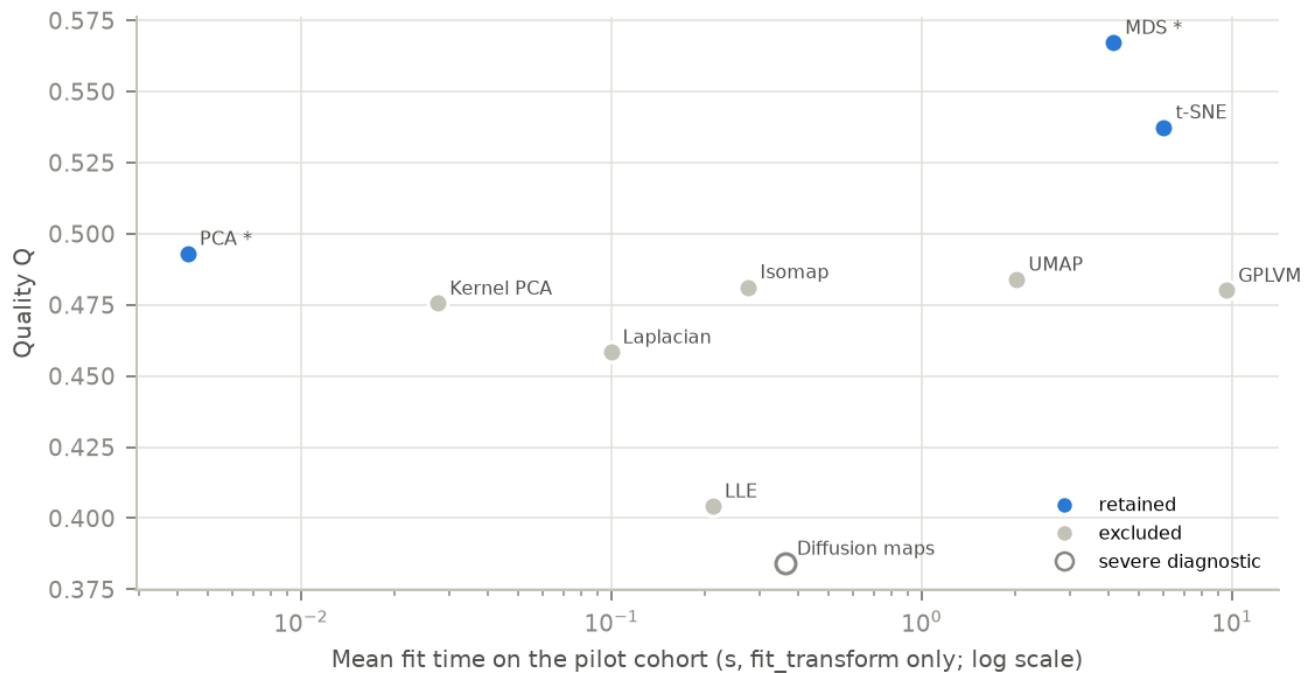
Highlighted rows were retained. Fit s is the mean fit\_transform wall time on the full pilot cohort. Seed S is the overlap between repeated seeds on the same cohort; it is shown for stochastic methods only.



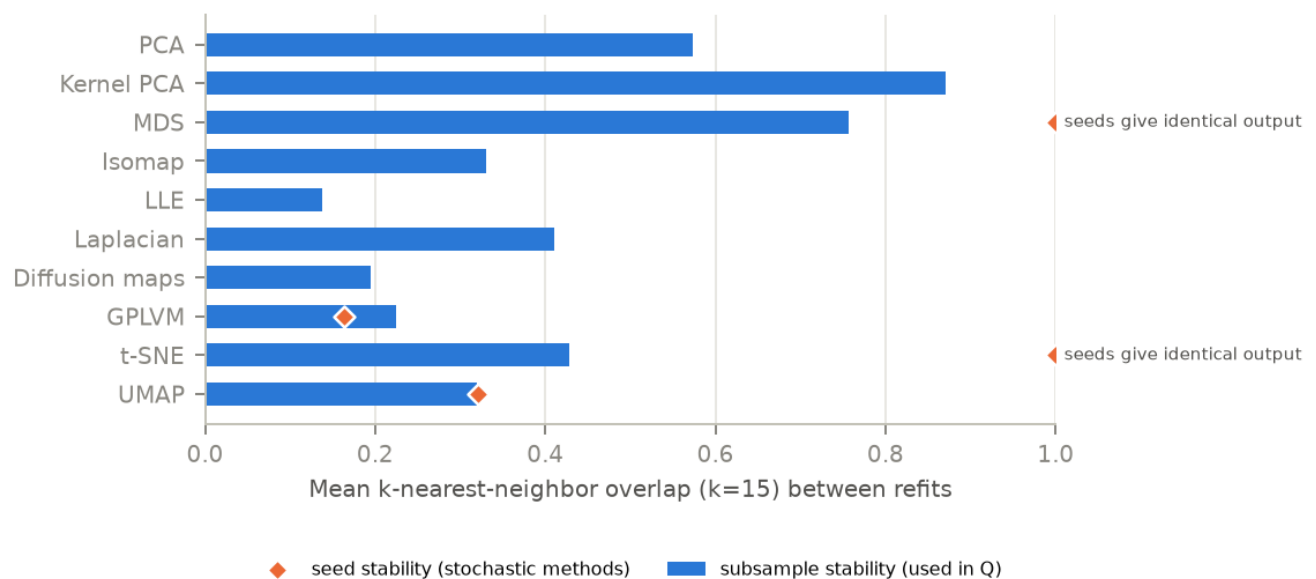
All ten pilot embeddings on the identical input (seed-774 fit). Titles give the pre-registered quality Q, the collapse ratio and the decision; colors are secondary labels only.



Label-free pilot components on the common cohort; each panel's axis spans the observed range.  $Q = 0.35 T + 0.35 R + 0.20 G + 0.10 S$ .



Fit-only runtime against quality. \* marks the epsilon-Pareto frontier (a method is redundant only if another is at least as good and materially better or faster; see the decision rules).



Subsample stability compares refits on overlapping 80% subsamples and applies to every method; seed stability compares repeated seeds on the same cohort.

## 5 • Evidence-based selection

Best valid pilot quality is 0.5672 (MDS); the 90% threshold is 0.5105.

A method is eligible if it meets that threshold, lies within the practical-equivalence margin of the best, or sits on the  $\epsilon$ -Pareto frontier. The margin is  $\max(2 \times \text{paired subsample SE}, 0.005)$ . On the  $\epsilon$ -Pareto frontier, a runtime difference counts only when it is  $\geq 2\times$  and  $\geq 1$  s.

PCA is retained as the linear baseline, with at most 4 candidates in total.

| Method     | Prior (not evidence)                          | Decision | Measured reason                                                                                                                                                                                |
|------------|-----------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PCA        | Linear variance baseline.                     | retain   | Q=0.4930 vs best mds 0.5672; 90% threshold 0.5105; equivalence margin 0.0088; fit runtime 0.00s. Retained as the linear baseline.                                                              |
| Kernel PCA | Nonlinear kernel structure; quadratic memory. | exclude  | Q=0.4755 vs best mds 0.5672; 90% threshold 0.5105; equivalence margin 0.0050; fit runtime 0.03s. Excluded: below threshold, outside the equivalence margin, and made redundant by pca, isomap. |

| Method         | Prior (not evidence)                                                             | Decision | Measured reason                                                                                                                                                                                            |
|----------------|----------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MDS            | Global pairwise geometry; iterative quadratic distance work.                     | retain   | Q=0.5672 vs best mds 0.5672; 90% threshold 0.5105; equivalence margin 0.0050; fit runtime 4.14s. Retained: meets the 90% threshold, practically equivalent to the best, on the quality-runtime frontier.   |
| Isomap         | Geodesic geometry conditional on a connected neighborhood graph.                 | exclude  | Q=0.4813 vs best mds 0.5672; 90% threshold 0.5105; equivalence margin 0.0050; fit runtime 0.28s. Excluded: below threshold, outside the equivalence margin, and made redundant by pca.                     |
| LLE            | Local affine reconstruction conditional on well-conditioned neighborhoods.       | exclude  | Q=0.4042 vs best mds 0.5672; 90% threshold 0.5105; equivalence margin 0.0063; fit runtime 0.21s. Excluded: below threshold, outside the equivalence margin, and made redundant by pca, kernel_pca, isomap. |
| Laplacian      | Local graph geometry conditional on connectivity.                                | exclude  | Q=0.4584 vs best mds 0.5672; 90% threshold 0.5105; equivalence margin 0.0050; fit runtime 0.10s. Excluded: below threshold, outside the equivalence margin, and made redundant by pca, kernel_pca, isomap. |
| Diffusion maps | Density-corrected diffusion geometry.                                            | exclude  | Excluded: severe diagnostic — collapsed layout (median collapse ratio $0.009 < 0.05$ and $< 0.25 \times \text{PCA } 0.253$ ); near-disconnected diffusion graph (Markov spectral gap $9.0\text{e-}08$ ).   |
| GPLVM          | Probabilistic nonlinear latent-variable model under a fixed optimization budget. | exclude  | Q=0.4800 vs best mds 0.5672; 90% threshold 0.5105; equivalence margin 0.0050; fit runtime 9.58s. Excluded: below threshold, outside the equivalence margin, and made redundant by pca, mds, isomap.        |
| t-SNE          | Stochastic local-neighborhood visualization.                                     | retain   | Q=0.5374 vs best mds 0.5672; 90% threshold 0.5105; equivalence margin 0.0050; fit runtime 6.01s. Retained: meets the 90% threshold.                                                                        |
| UMAP           | Stochastic neighborhood-graph embedding.                                         | exclude  | Q=0.4839 vs best mds 0.5672; 90% threshold 0.5105; equivalence margin 0.0050; fit runtime 2.02s. Excluded: below threshold, outside the equivalence margin, and made redundant by pca.                     |

## Robustness of the shortlist

Each row below re-runs the same selection mechanics under one rule change on the same saved evidence; only the first row determined the analysis.

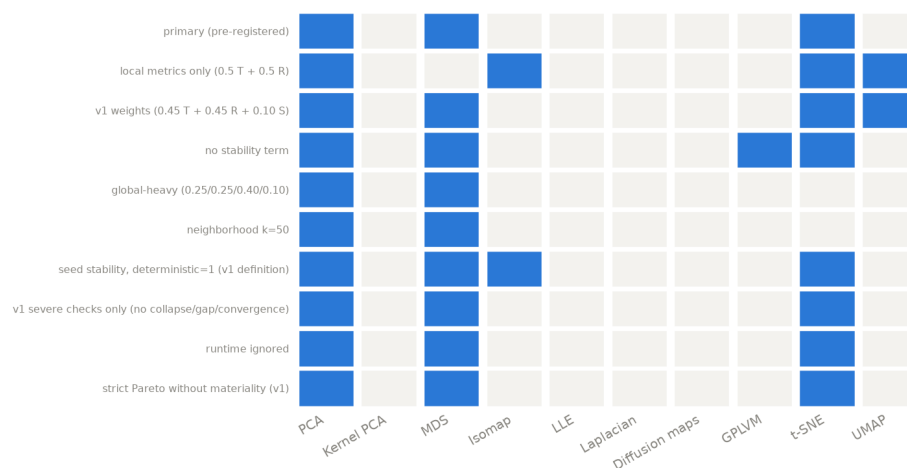
| Rule variant                                        | Shortlist                |
|-----------------------------------------------------|--------------------------|
| primary (pre-registered)                            | PCA, MDS, t-SNE          |
| local metrics only (0.5 T + 0.5 R)                  | PCA, t-SNE, UMAP, Isomap |
| v1 weights (0.45 T + 0.45 R + 0.10 S)               | PCA, t-SNE, MDS, UMAP    |
| no stability term                                   | PCA, t-SNE, MDS, GPLVM   |
| global-heavy (0.25/0.25/0.40/0.10)                  | PCA, MDS                 |
| neighborhood k=50                                   | PCA, MDS                 |
| seed stability, deterministic=1 (v1 definition)     | PCA, t-SNE, MDS, Isomap  |
| v1 severe checks only (no collapse/gap/convergence) | PCA, MDS, t-SNE          |
| runtime ignored                                     | PCA, MDS, t-SNE          |
| strict Pareto without materiality (v1)              | PCA, MDS, t-SNE          |

Selection frequency across variants: PCA 100% · MDS 90% · Isomap 20% · GPLVM 10% · t-SNE 80% · UMAP 20%.

## From the v1 rules to the pre-registered v2 rules

This cumulative ablation is computed on the same evidence; each step changes one rule.

| Step                                              | Shortlist               |
|---------------------------------------------------|-------------------------|
| A0 v1 rules on v2 evidence (fit-only timing)      | PCA, t-SNE, MDS, Isomap |
| A1 + runtime materiality (epsilon-Pareto)         | PCA, t-SNE, Isomap      |
| A2 + collapse / spectral-gap / convergence checks | PCA, t-SNE, Isomap      |
| A3 + global-structure component                   | PCA, t-SNE, MDS, Isomap |
| A4 + subsample stability replaces seed stability  | PCA, MDS, t-SNE         |
| A5 + equivalence-to-best eligibility (= primary)  | PCA, MDS, t-SNE         |



Shortlist under alternative rule variants on the same saved pilot evidence (filled = retained). Only the first row determined the analysis; the others test robustness.

## 6 • Shortlist refinement

Only retained methods were tuned. Each grid point was evaluated with the same seeds, subsamples and quality as the pilot. A non-default point replaces the default only if it gains more than the paired equivalence margin.

| Method | Grid | Changed parameters    | Status  | Q     | S     | Fit s |
|--------|------|-----------------------|---------|-------|-------|-------|
| PCA    | 0    | default               | success | 0.493 | 0.574 | 0.00  |
| MDS    | 0    | default               | success | 0.567 | 0.756 | 4.13  |
| MDS    | 1    | init=random; n_init=4 | success | 0.528 | 0.448 | 56.09 |
| t-SNE  | 0    | default               | success | 0.537 | 0.428 | 6.15  |
| t-SNE  | 1    | perplexity=15         | success | 0.531 | 0.393 | 4.56  |
| t-SNE  | 2    | perplexity=50         | success | 0.536 | 0.476 | 7.26  |

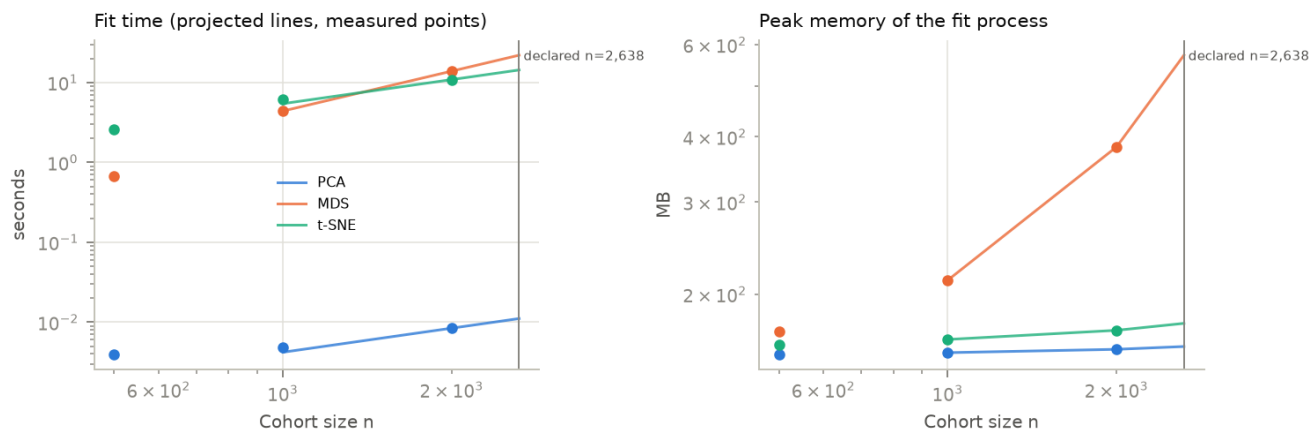
- **PCA**: default has the highest quality.
- **MDS**: default has the highest quality.
- **t-SNE**: default has the highest quality.

## 7 • Final cohort size

Shortlisted methods were fitted on stratified probe cohorts of 500, 1000, 2000 observations. Log-log slopes (floored at 1) projected their cost at each candidate size, with a 2× safety factor. The limits were: per-run timeout 30 min, final-stage budget 40 min, memory 24 GB.

**Declared final size: 2,638** of 2,638 available observations (cap 3,000); binding constraint: all available observations.

| Method | Time exponent | Memory exponent | Measured at n=2,000 | Projected at declared n |
|--------|---------------|-----------------|---------------------|-------------------------|
| MDS    | 1.66          | 2.31            | 13.88 s · 383 MB    | 22.0 s · 573 MB         |
| PCA    | 0.82          | 1.65            | 0.01 s · 157 MB     | 0.0 s · 159 MB          |
| t-SNE  | 0.84          | 1.48            | 10.88 s · 171 MB    | 14.3 s · 176 MB         |



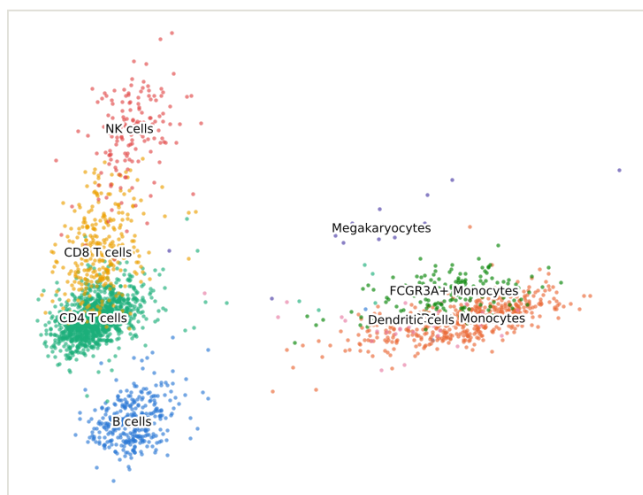
Scaling probe of the shortlisted methods and the conservative projections used to declare the final cohort size (all available observations).

## 8 · Final results

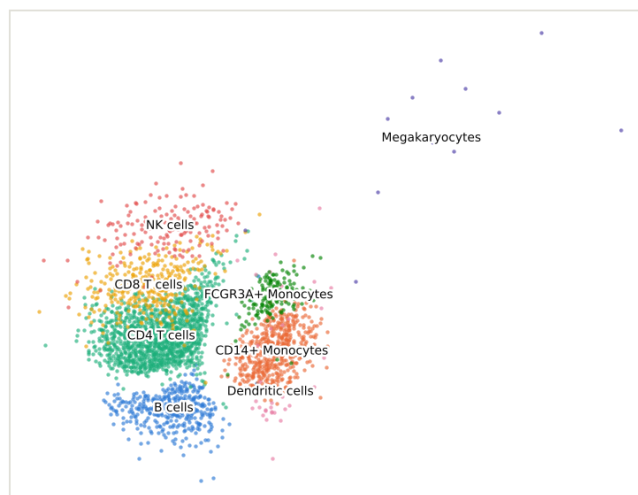
| Method | T     | R     | G     | T (k=50) | R (k=50) | Collapse | Silhouette (secondary) | Fit s |
|--------|-------|-------|-------|----------|----------|----------|------------------------|-------|
| PCA    | 0.825 | 0.051 | 0.808 | 0.833    | 0.128    | 0.265    | 0.404                  | 0.01  |
| MDS    | 0.848 | 0.064 | 0.842 | 0.854    | 0.154    | 0.454    | 0.282                  | 32.57 |
| t-SNE  | 0.897 | 0.208 | 0.654 | 0.882    | 0.249    | 0.488    | 0.419                  | 14.90 |

Final metrics are in-cohort descriptions of the declared final cohort. They are not held-out estimates and should not be compared with pilot values as if the observation sets were identical. Chosen parameters: PCA (svd\_solver=full); MDS (init=classical\_mds; max\_iter=1000; metric=euclidean; metric\_mds=True; n\_init=1; n\_jobs=1; normalized\_stress=auto); t-SNE (init=pca; learning\_rate=auto; max\_iter=750; n\_jobs=1; perplexity=30).

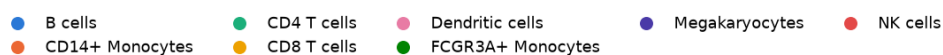
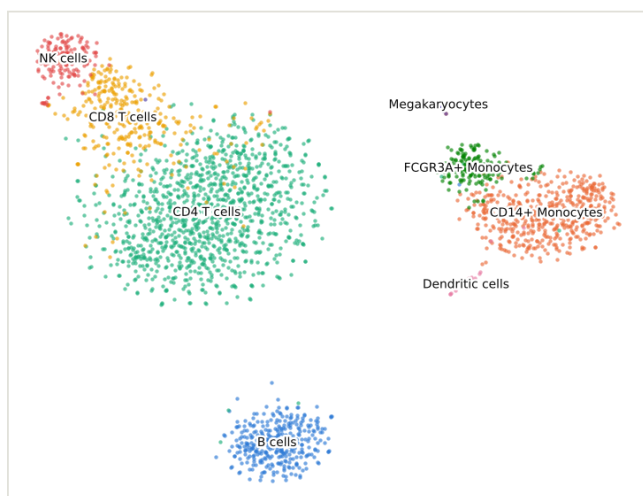
PCA T 0.825 · R 0.051 · G 0.808



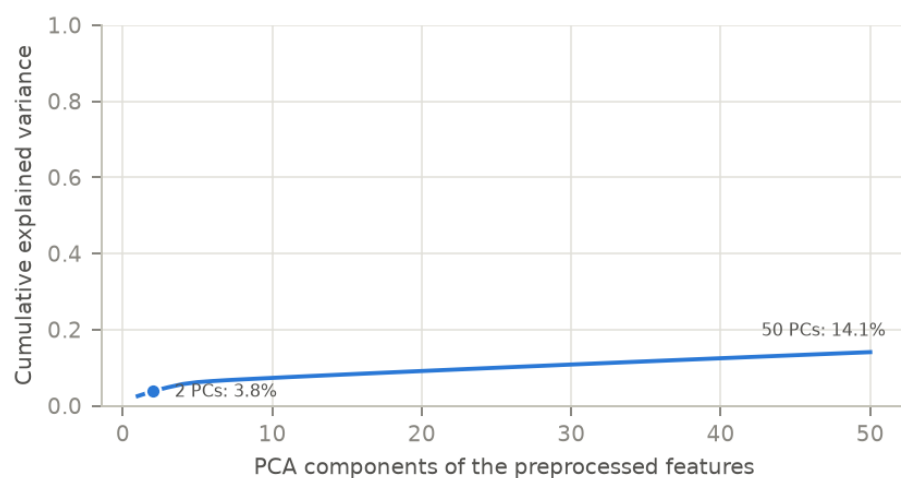
MDS T 0.848 · R 0.064 · G 0.842



t-SNE T 0.897 · R 0.208 · G 0.654



Final embeddings of the 3 shortlisted methods on the declared final cohort (n=2,638). Class names sit at class medians; colors are secondary labels, and visual separation is not evidence of biological or histological truth.



Variance of the preprocessed feature space retained by the common PCA representation (final cohort).

## 9 • Limitations

- Q is one pre-registered weighting of label-free criteria. The robustness table shows which conclusions survive other reasonable weightings; no weighting establishes a universally superior method.

- The collapse and spectral-gap checks are conservative and catch only extreme failures. A layout can still be visually degenerate above the thresholds.
- Runtimes are single-threaded CPU measurements on one node, and material-difference tolerances absorb run-to-run noise. The size projections are conservative planning envelopes, not fitted complexity laws.
- GPLVM uses a fixed epoch budget (loss plateau recorded), not a convergence proof.
- PBMC annotations come from a prior clustering, and PathMNIST pixel distances need not reflect histological semantics. Class colors and silhouettes are secondary diagnostics, not validation.

## 10 • Reproducibility

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Configuration SHA-256 d393805aba3bd17ddcd2001087cd06ff58176ea111d205bdf55fec0488026f8 · code SHA-256 c9dc56a74a94fb64c58f5b4e318e267e9ff66183c325f51e8558b23aaf0dd1 · git commit 766f1d17e25a6cd1bdec4dbf62c61140a32e1d3c · job 2840720 on c0317.ll.unc.edu · 2026-09-29T02:03:19.266091+00:00.

Software: numpy 2.4.6 · pandas 2.3.3 · scipy 1.17.1 · scikit-learn 1.9.1 · anndata 0.12.19 · scanpy 1.11.5 · umap-learn 0.5.12 · pydiffmap 0.2.0.1 · medmnist 3.0.2 · torch 2.14.0+cpu · gpytorch 1.15.2 · matplotlib 3.11.2 · weasyprint 70.0 · pypdf 6.19.0 · openai 3.18.0.

Source: <https://scanpy.readthedocs.io/en/latest/api/generated/scanpy.datasets.pbmc3k.html> · accessed 2026-09-29. Cached file checksums: pbmc3k\_processed.h5ad 0db367b991dd...; pbmc3k\_raw.h5ad 89a96f1beaa2...

References: Scanpy PBMC3k documentation and 10x Genomics PBMC3k; Yang et al., MedMNIST v2, Scientific Data 10:41 (2023); Titsias & Lawrence, Bayesian GPLVM, AISTATS (2010); van der Maaten & Hinton, t-SNE, JMLR (2008); McInnes et al., UMAP (2018); Coifman & Lafon, Diffusion maps (2006); Venna & Kaski, trustworthiness (2001). Full links: docs/METHOD\_NOTES.md and docs/DATA\_PROVENANCE.md.